A Numerical Algorithm for simulating multiphase flows: ice, melt, vapor, deformable solids, high speed, and low speed flows *

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ABSTRACT: A new Eulerian method is presented for simulating complex multiphase flows consisting of any number of types of materials, to include compressible fluids, incompressible fluids, ice, melt, vapor, and solids undergoing plastic deformation. The interfaces between materials is modeled and discretized as sharp.

1 Mathematical Modeling

There are M materials and each material $1 \leq m \leq M$ occupies the region,

$$\Omega_m(t) = \{(\vec{x}, t) | \phi_m(\vec{x}, t) \geq 0\} \tag{1}$$

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The $\phi_m(\boldsymbol{x}, t)$ are level set functions[2, 1]. The zero level set of such functions demarcate the boundary of material m:

$$\partial\Omega_m(t) = \{(\vec{x}, t))|\phi_m(\vec{x}, t) = 0\} \tag{2}$$

In the bulk regions of material m we have the following equations for compressible materials:

$$\rho_t + \nabla \cdot (\rho\vec{u}) = 0 \tag{3}$$

$$(\rho\vec{u})_t + \nabla \cdot (\rho\vec{u} \otimes \vec{u} + p\mathbb{I} - \tau) = \vec{F}_{body} \tag{4}$$

$$(\rho E)_t + \nabla \cdot ((\rho E + p)\vec{u} - \vec{u} \cdot \tau - k\nabla T) = \vec{u} \cdot \vec{F}_{body} \tag{5}$$

$$(\rho Y)_t + \nabla \cdot (\rho Y \vec{u}) = \nabla \cdot (\rho D \nabla Y) \tag{6}$$

$$\tau = 2\mu\bar{D} \tag{7}$$

$$\bar{D} = D - \frac{\mathrm{Trace}(D)\mathbb{I}}{\mathrm{Trace}(\mathbb{I})} \tag{8}$$

$$D = \frac{\nabla\vec{u} + (\nabla\vec{u})^T}{2} \tag{9}$$

$$\rho E = \rho e_{\mathrm{int}} + \frac{1}{2}\rho|\vec{u}|^2 \tag{10}$$

$$p = p(\rho, e) \quad e = c_v T \tag{11}$$

If the material is a viscoelastic FENE-CR material, then one has the additional equation for the configuration tensor A,

$$A_t + \vec{u} \cdot \nabla A = A \cdot (\nabla\vec{u})^T + \nabla\vec{u} \cdot A - \frac{f(R)}{\lambda}(A - \mathbb{I}), \tag{12}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + \frac{\tilde{\eta}_P}{\lambda} f(R) A. \tag{13}$$

If the material is an elastic material undergoing plastic deformation, then one has the following additional equation for the deviatoric stress tensor,

$$s_t + \vec{u} \cdot \nabla s + s\Omega - \Omega s = 2G(\bar{D} - D^p), \tag{14}$$

and the momentum constitutive law term is modified,

$$\tau = \tau + s. \tag{15}$$

If the material is an elastic material undergoing plastic deformation, then the plastic strain equation is solved too,

$$\epsilon_t^p + \vec{u} \cdot \nabla \epsilon^p = \sqrt{\frac{2}{3}} |\Lambda|. \tag{16}$$

$$|\Lambda| = \sqrt{\frac{3}{2}} \sqrt{D^p : D^p} \tag{17}$$

$$D^p = \Lambda N \quad N = \sqrt{3/2} \frac{s}{S_e} \quad (\dot{\epsilon}^p)^2 = \frac{2}{3} \Lambda^2 \tag{18}$$

Ω is the spin tensor,

$$\Omega_{ij} = \frac{1}{2} (\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}) \tag{19}$$

The effective stress is defined as,

$$S_e = \sqrt{\frac{3}{2}} \sqrt{s : s} \tag{20}$$

The Johnson-Cook model yield stress is given by,

$$\sigma_y = (A + B(\epsilon^p)^n)(1 + C \ln(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0}))(1 - \theta^m) \quad \theta = \frac{T - T_0}{T_m - T_0} \tag{21}$$

One form of the Zerilli Armstrong model yield stress is given by,

$$\beta = \beta_0 - \beta_1 \log(\dot{\epsilon}^p / \dot{\epsilon}^p_{ref}) \tag{22}$$

$$\sigma_y = A + B(\epsilon^p)^n e^{-\beta T} \tag{23}$$

The energy contribution due to elasto-plastic deformation is modeled as,

$$\beta \dot{\epsilon}^p S_e, \tag{24}$$

and is added to (5). In order to determine Λ, as it appears in (18), the following "radial return algorithm" is implemented (see [3] for the case when $n = 1$ in (21) or (23)):

1. $\dot{\epsilon}^p = 0$, $k = 0$

2. $S_e = \sqrt{\frac{3}{2}} \sqrt{s : s}$

3. $\sigma_y = \sigma_y(T, \epsilon^p, \dot{\epsilon}^p)$

4. if $S_e - \sigma_y \leq 0$ then $\Lambda = 0$ (i.e. $D^p : D^p = 0$).

5. if $S_e > \sigma_y$ then,

6. Solve the following equation for ϵ^p_{new}:

$$S_e - \sigma_y = 3G(\epsilon^p_{new} - \epsilon^p) + B((\epsilon^p_{new})^n - (\epsilon^p)^n) \tag{25}$$

7.

$$\dot{\epsilon}^p = \frac{\epsilon^p_{new} - \epsilon^p}{\Delta t} \tag{26}$$

8.

$$\Lambda = \sqrt{\frac{3}{2}}\dot{\epsilon}^p \tag{27}$$

$$D^p = \Lambda N \tag{28}$$

9. $k = k + 1$

10. if $k = 1$ then go back to step 3.

The mathematical model at sharp interfaces is given as follows. First, define a "closest point" function:

$$\vec{x}_c(\vec{x}) = \text{argmin}_{\vec{y},m} \{||\vec{y} - \vec{x}|| | \phi_m(\vec{y}) = 0\} \tag{29}$$

Taking into account mass transfer, a source material's level set function (e.g. ice that is melting or liquid that is turning into vapor) satisfies

$$(\phi_{m_s})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_s}| \tag{30}$$

The destination material's level set function satisfies

$$(\phi_{m_d})_t + \vec{u}_{m_s}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_s}}|\nabla\phi_{m_d}| \tag{31}$$

The above equations are equivalent to the following:

$$(\phi_{m_s})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_s} = -\frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_s}| \tag{32}$$

$$(\phi_{m_d})_t + \vec{u}_{m_d}(\vec{x}_c(\vec{x})) \cdot \nabla\phi_{m_d} = \frac{|\dot{m}(\vec{x}_c(\vec{x}))|}{\rho_{m_d}}|\nabla\phi_{m_d}| \tag{33}$$

Now, we define the interface normal:

$$\vec{n}_m = \frac{\nabla \phi_m}{||\nabla \phi_m||} \tag{34}$$

The interface curvature is,

$$\kappa_m = \nabla \cdot \vec{n}_m \tag{35}$$

The rate of mass transfer is,

$$\dot{m} = \frac{k_{m_s} \nabla T_{m_s} - k_{m_d} \nabla T_{m_d}}{L} \cdot \vec{n}_{m_d}. \tag{36}$$

The phase change interface temperature condition is,

$$T_{m_s} = T_{m_d} = T_{\text{saturation}}(P_{\text{equilibrium}}) \tag{37}$$

For evaporation, the following rate of mass transfer should match that in (36):

$$\dot{m} = \frac{-\vec{n}_{m_s} \cdot \nabla Y_{m_d} \rho_{m_d} D}{Y^{\Gamma} - 1} \tag{38}$$

Let the tensor σ represent pressure, viscosity, viscoelastic, and elastic contributions to the momentum equations. One has:

$$\sigma = -p\mathbb{I} + \tau. \tag{39}$$

The stress jump conditions are

$$\vec{n}_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = -\gamma_{m_s,m_d} \kappa_{m_s} - \dot{m}^2 (\frac{1}{\rho_{m_s}} - \frac{1}{\rho_{m_d}}) \tag{40}$$

$$(\vec{t}_{i,tangent})_{m_s} \cdot (-\sigma_{m_s} + \sigma_{m_d}) \cdot \vec{n}_{m_s} = 0 \quad i = 1, 2 \tag{41}$$

The velocity jump conditions are

$$(\vec{u}_{m_s} - \vec{u}_{m_d}) \cdot \vec{n}_{m_s} = \dot{m} (\frac{1}{\rho_{m_d}} - \frac{1}{\rho_{m_s}}) \tag{42}$$

2 budget

The budget should be arount 495K for this phase I grant.

1. Sussman; two months (28K)

2. Shoele; ?

3. Arienti (100K)

4. graduate students

5. postdoc

3 Applications

1. Jet in cross flow problem (military jet engines, Pratt and Whitney)

2. Diesel injectors

3. Rotating Detonation Engines (see latest article with Everett Wenzel and Marco Arienti).

4 Proposed algorithm

0. Reseed particles with dense particle regions in narrow bands about interfaces (but not on interfaces!). Initialize the state of the particles using the associated PINN from the previous time step.

1. Cell integrated semi Lagrangian advection and particle advection. Particle advection uses PINN velocity.

2. Cell integrated semi Lagrangian advection due to phase change.

3. Viscosity, Viscoelastic, and elastic forces.

4. Temperature Diffusion

5. Mass Fraction Diffusion

6. Pressure Gradient

7. Identify all separate contiguous material regions (this is done already, see Wang, Simakhina, Sussman, or Sussman Ohta)

8. For each separate region, train a NN using PINN, CFD, and experimental data. The CFD data on the interface of each material region has the most weight. The PINN is trained in a Lagrangian frame of reference. x, t^{n+1} mapped to itself in the NN. $x - u\Delta t, t^n$ mapped to x, t^n in the NN. The NN is valid for $t^n < t < t^{n+1}$. It could be that trained NN can be recycled if the associated material regions are topologically similar.

9. Go back to step 0.

References

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